Problem 13

Evaluate the integral:

$$\int (2^x + 5^x)^2 \, dx.$$

Solution

To solve the given integral $\int (2^x + 5^x)^2 \, dx$, we start by expanding the square of the expression inside the integral.

Step 1: Expand the square

Using the algebraic identity $(a+b)^2 = a^2 + 2ab + b^2$, the expression $(2^x + 5^x)^2$ expands to:

$$(2^x + 5^x)^2 = (2^x)^2 + 2(2^x)(5^x) + (5^x)^2.$$

Simplify each term:

$$(2^x)^2 = 2^{2x}, \quad 2(2^x)(5^x) = 2 \cdot 2^x \cdot 5^x = 2 \cdot (2 \cdot 5)^x = 2 \cdot 10^x, \quad (5^x)^2 = 5^{2x}.$$

Thus, the integrand becomes:

$$(2^x + 5^x)^2 = 2^{2x} + 2 \cdot 10^x + 5^{2x}.$$

Step 2: Rewrite the integral

The integral can now be written as:

$$\int (2^x + 5^x)^2 \, dx = \int 2^{2x} \, dx + \int 2 \cdot 10^x \, dx + \int 5^{2x} \, dx.$$

Step 3: Solve each term

We evaluate each integral separately using the formula for the integral of an exponential function:

$$\int a^x \, dx = \frac{a^x}{\ln(a)} + C, \quad \text{for } a > 0,\, a \neq 1.$$

1. **First term: $\int 2^{2x} \, dx$:** Let $a = 2^{2x}$. We rewrite 2^{2x} as $(2^2)^x = 4^x$. Then:

$$\int 2^{2x} \, dx = \int 4^x \, dx = \frac{4^x}{\ln(4)}.$$

2. **Second term: $\int 2 \cdot 10^x \, dx$:** Using the constant multiple rule, $\int 2 \cdot 10^x \, dx = 2 \int 10^x \, dx$. Solve the integral:

$$\int 10^x \, dx = \frac{10^x}{\ln(10)}.$$

So:

$$\int 2 \cdot 10^x \, dx = 2 \cdot \frac{10^x}{\ln(10)} = \frac{2 \cdot 10^x}{\ln(10)}.$$

3. **Third term: $\int 5^{2x} \, dx$:** Let $a = 5^{2x}$. Rewrite 5^{2x} as $(5^2)^x = 25^x$. Then:

$$\int 5^{2x} \, dx = \int 25^x \, dx = \frac{25^x}{\ln(25)}.$$

Step 4: Combine the results

Adding the results of the three terms, we have:

$$\int (2^x + 5^x)^2 \, dx = \frac{4^x}{\ln(4)} + \frac{2 \cdot 10^x}{\ln(10)} + \frac{25^x}{\ln(25)} + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$\int (2^x + 5^x)^2 \, dx = \frac{4^x}{\ln(4)} + \frac{2 \cdot 10^x}{\ln(10)} + \frac{25^x}{\ln(25)} + C.$$